

Ryme

GAS ANALYZER

Maintenance manual
Version 01

En

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1 Introduction.

Thank you for purchasing the unit of measurement of gases RYME included in the gas analyzer RY-500 AH. With a newly designed, it benefits from the most advanced technologies both software and hardware level to bring them to serve their priorities: speed test, accuracy, reliability and low maintenance. Especially suited to workshops, its easy use allows quick and effective contact with all the possibilities offered by this device. We are therefore convinced that will give total satisfaction.

This manual describes how to use the unit of measurement of gases from the same installation to measurement procedures, through measures to be taken in case of failure. Please read this manual carefully before using the appliance in order to take best advantage. Keep it in a safe place for future reference if needed.

Comment:

The information contained in this document may change without notice.

In any case RYME be liable for direct or indirect damages, or for any losses or expenses resulting from non-conforming use.

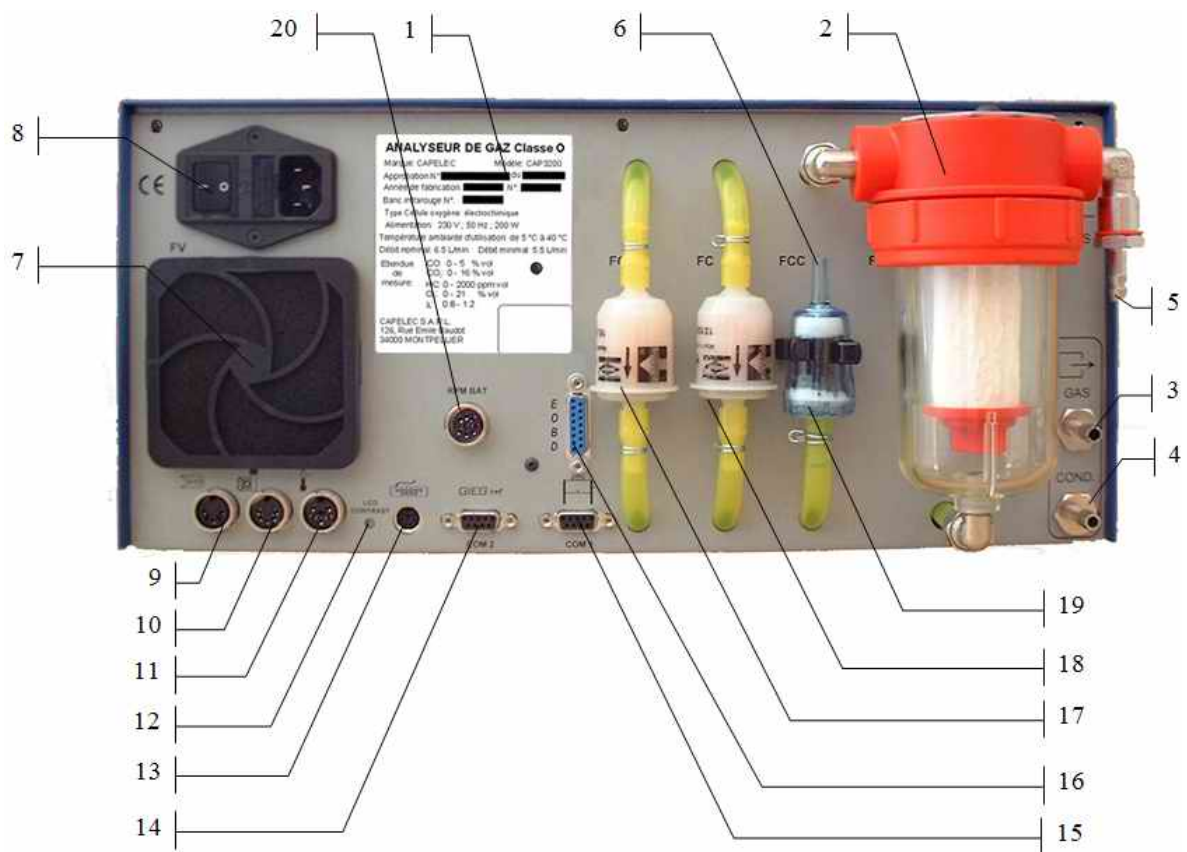
2 Unit.

The unit of measurement of gases has an LCD screen, a 5-key keyboard, a printer and a gas analysis module for gasoline vehicles. The measuring cell is connected to the measuring unit via a communication cable and power.

2.1. Gas analyzer unit, from the front.

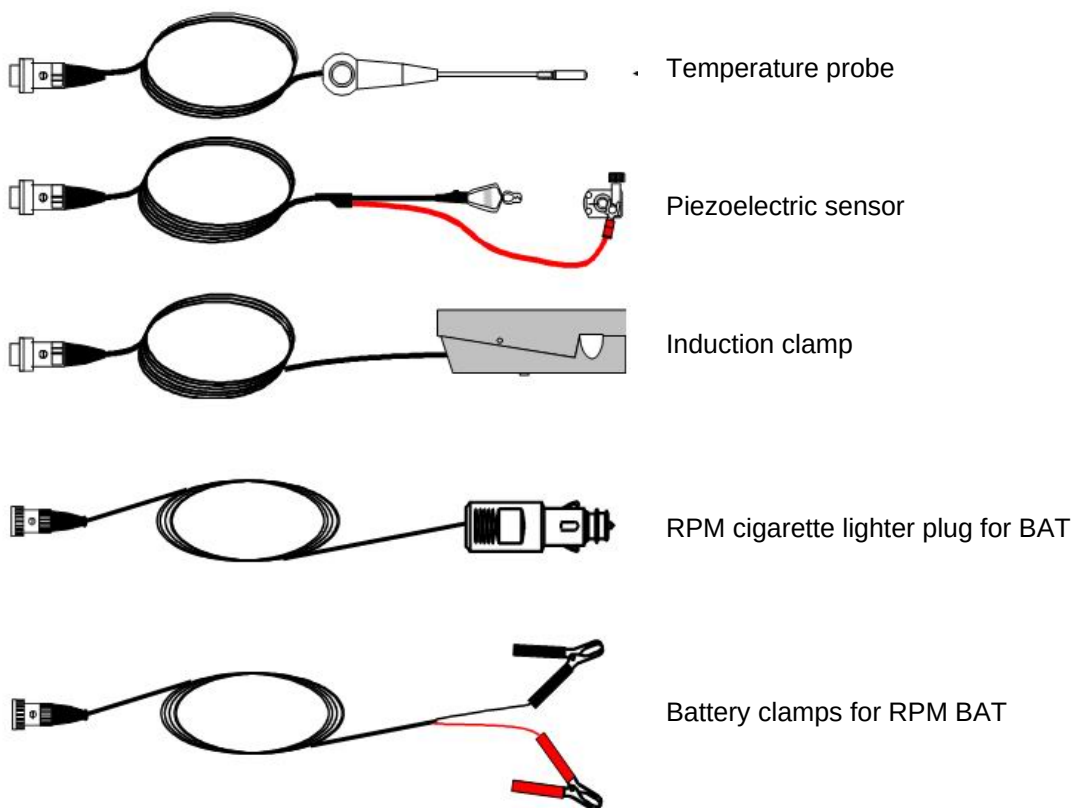


2.2. Gas analyzer unit, from the back.



1	Nameplate
2	Filter decanter (DE)
3	Output of measured gas (GAS)
4	Condensate outlet (COND)
5	Inlet gas to be measured (GAS)
6	Check ambient air to zero
7	Fan filter (FV)
8	Mains filter with fuse, switch 230V
9	Induction clamp input
10	Check the piezoelectric sensor and TTL
11	Check the oil temperature sensor
12	Contrast LCD
13	PS2 keyboard input
14	COM2 serial port
15	COM1 serial port
16	EOBD
17	Protective filter pump gas (FG)
18	Protection filter condensate pump (FC)
19	Activated carbon filter (FCC)
20	Check engine speed with battery.

2.3. Temperature module and tachometer.



3 Maintenance and security

3.1. Overview

Each unit of measurement of gases has been checked before delivery, a representative of the Department of measuring instruments that puts the seal of initial verification.

The seals prohibiting access to physical elements that may have an influence on the metrological quality of results can only be removed by an authorized person. Ensure the compliance of material. Any action on the unit of measurement or the cell, which requires the withdrawal of these seals can only be made by an authorized person.

3.2. Cleaning and surface wires

The unit of measurement, display, keyboard, and cables can be cleaned with alcohol on a clean cloth. The use of any solvent is strongly discouraged.

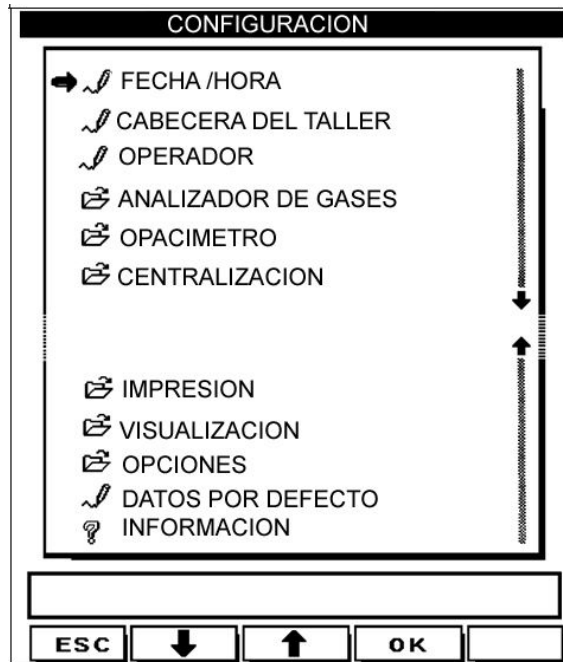
The screen should be cleaned with cotton without abrasive materials.

4 Settings.

4.1. Overview.

To run the Setup menu, select in the main "Settings" menu online.

The following screen appears:



This is a menu composed of several propositions.

Each proposition allows access to a configuration window.

All these windows are described below.

4.2. Date and hour.

Update the time and date of the measurement unit.



You have to select the field with OK, to change the value you must press the UP and DOWN arrows. Finally, to save the new settings, you must click OK.

Out by the ESC key.

4.3. Workshop heading.

This application lets you enter and customize the information related to the workshop. This information is printed at the beginning of the test report. The window for entering the data is presented as follows.



To enter data should be used:

- Or five keys at the bottom of the screen; in the event that the keyboard option is not installed or is nonexistent. The UP and DOWN arrows allow rapid movement of numbers and figures. The >> button switches to the next character and instead << return to the previous character. Finally, OK lets change line to the last line. Following this line, the program exits and saves the information. Para teclear los datos se debe utilizar:
- • Either the alphanumeric keypad supplied as an option. The UP, DOWN (optional keyboard) will cycle line. To delete a character, use the key ...← We occupy the top right of the keyboard, or LEFT arrow. To move to the next phase only need to press the OK key (keyboard of 5 keys) or press INPUT.



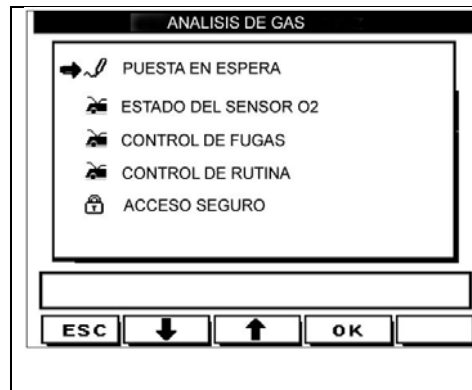
The Print button to print in thermal printer, the header workshop as it will appear for a test report.



This key lets you enlarge and dwarf characters.

5 Gas analysis.

Allows access to a menu to choose the time power standby pumps; also reveals the state of the oxygen sensor, carry out an escape, see the control routine and finally access the Security menu.



5.1. On hold

To put on hold, select the "put on hold" tab then press OK. The time remaining before the party gas analysis to standby appears in the dialog box. With the UP and DOWN arrow can set the time to suit your needs. press OK to confirm your choice.

5.2. O2 sensor status

This window reports the status of the oxygen sensor. The measurement unit performs a consensus diagnosis and reports its wear. The wear is graduated from 0 to 100%, with 100% more satisfying to the state of wear.

5.3. Leak control

The Leak Test is a procedure that allows the detection of any leakage problems in the pneumatic circuit. This control must be made leakage daily basis. The unit of measurement ask user for a leak test only turn it on and automatically every four hours. The presence of a leak involves the immediate action of maintenance.

This test is fully guided. The operator must follow carefully the instructions on the screen.

- Test upstream of the pump:

The message "obstructs the entry" appears. The sealing of the entry is automatically detected by the measurement unit. The pump is then stopped and the following message appears: "Test in Progress". You need to keep the sealed entrance. This phase lasts 11 seconds maximum. At the end of this phase, the diagnosis "Presence of escape" or "No leak" is indicated.

Press **OK** to continue the test. Otherwise, **ESC**.

- Test pump downstream:

The following message appears "Obstruct exit". The shutter output is automatically detected by the measurement unit. The pump is then stopped and the following message appears: "Test in Progress". You need to keep the sealed entrance. This phase lasts 8 seconds maximum. At the end of this phase, the diagnosis "Presence of escape" or "No leak" is indicated.

Stop the test and exit **ESC** :

The presence of leakage measurement blocks until it has made a new test with a final result of type "No leak".

5.4. Routine control.

This window consists of 2 pages (PG1 and PG2). Page 1 (PG1) reports in real time gas pressure, ambient temperature, lights and detectors. Indicates the computer BENCH version and serial number. And finally, this page shows the status of gas analysis module (word_status). Page 2 (PG2) provides information on the date and names of participants, indicating the computer version of Part gas analysis measurement unit, provides the value of PEF and finally indicates the verification code program (checksum).

CONTROL DE RUTINA		CONTROL DE RUTINA	
Pression gaz: 1010 mbar Temp. ambiante: 29.3 °C Temp. lampe: 39.06 °C Temp. détecteur: 55.79 °C BENCH: V5.129 N°: 6433		Intervention(s): 29/08/2002 ale 29/08/2002 fdsd 29/08/2002 hjkhkjh 29/08/2002 hgiii 16/09/2002 ale 17/09/2002 uhk 17/09/2002 dgdfg 19/09/2002 hhs	
word_status(0)=0x4001 word_status(5)=0x0000 word_status(1)=0x0000 word_status(6)=0x0000 word_status(2)=0x0000 word_status(7)=0x0002 word_status(3)=0x0000 word_status(8)=0x0000 word_status(4)=0x0000 word_status(9)=0x000B		Unidad medida: V1.00 F.E.P. = 0.492 Checksum gaz: bdd8	
ESC PG1 PG2		ESC PG1 PG2	

5.5. Secure access.

Enter the secret code of 5 keys, then **OK** .

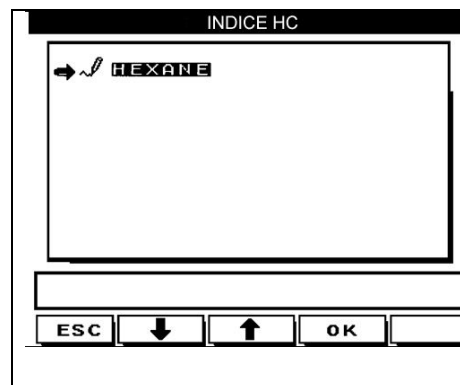
The system asks the user to enter the name "Made by:" This name must contain between 3 and 15 characters. This information is stored in memory and can not be deleted.

This part of the program is reserved for persons responsible for periodic inspection of the unit.



- HC index

Allows access a submenu indicating that the display of HC is equivalent hexane.



During calibration, the value of HTC is performed automatically on propane.

Notably, the apparatus displayed by the PEF PEF is propane to a value close to HC 2000 ppm vol.

- Calibration

The second menu allows calibration of the apparatus consists of several propositions:



- the adjustment procedure CO, HC, CO2:

→ Setting the calibration gas concentrations:

IMPORTANT: HC be made equivalent propane.

Recall that the gas cylinder required for setting the analyzer contains the following concentrations:

- CO: 3,50 % vol.
- CO₂: 14,0 % vol.
- HC: 2000 ppm vol. Propane
- Rest N₂

The operator must enter values for the concentrations of CO, CO₂ and HC gases. To move to the next stage you must press OK or INPUT.

→ Running the setup procedure :

It guides the user through the process by messages that tell the different stages that the unit of measurement performed for an adjustment procedure.

- O₂ initialization:

→ O₂/air:

To change the ratio O₂ / Air, select the tab and click OK, change the value and press OK to confirm.

Significantly, it is obligatory to leave the value at 20.90%. This is the value used during each reset the device.

→ New O₂ probe:

This tab is used to install a new O2 sensor. You must follow the procedure first asks to change the O2 sensor, then press OK. After this stage, the measurement unit performs the initialization of the sensor for 1

- Initialization NOx:

Ídem new O2 probe.

- AMB access

The latter MICROBENCH menu allows access through a computer via RS232. The operator can thus perform all operations set forth above.



When this window is activated, the user can communicate with the bank gas analysis.

Exit to return to measurement mode: **ESC**

6 Tachometer.

6.1. Working principle

The revolutions account is based on the frequency analysis of noise on the battery terminals. Some peripheral vehicle, such as the alternator, have a direct influence on the volta

A cable to the battery.

* A wire to the cigarette lighter socket. When the motor rotates, this is not perfectly continuous. Indeed, looking closely, the battery voltage undergoes small oscillations proportional to engine speed.

From the analysis of these disturbances, the counter will remove the motor to submit regime, using an analog or digital signal, the unit of measurement of gases..

6.2. Technical characteristics.

- * * Automatic initialization.
- * * Measuring ranges of measurement: 400-9999 rpm.
- * * Maximum error:
- * <40 r.p.m. for a lower engine speed to 2,000 r.p.m.
- * <2% in other cases.
- * * Response time: <1 second.
- * * Refresh rate measurement: 10 per second.
- * * Initialization time: <17 seconds

6.3. List of supplied items.

- * A cable to the battery.
- * A wire to the cigarette lighter socket.

6.4. Connection

- * Connecting to the cigarette lighter socket

It is the simplest solution. Lets have a measurement system without opening the cover of the vehicle. However, in some vehicles, the measurement system may be less successful if the connection is made directly to the battery.

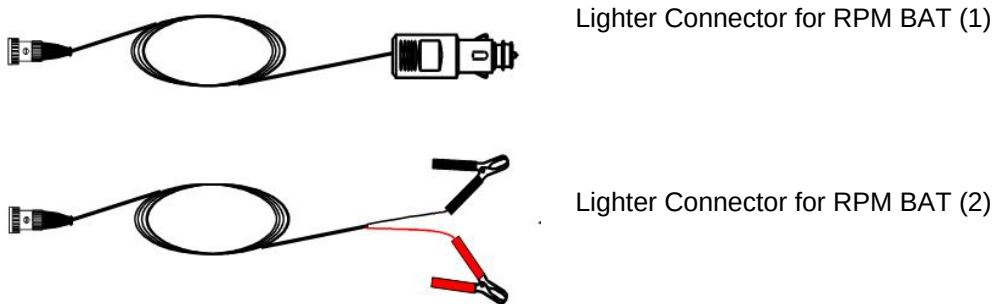
Connect the cable (1) on BAT input RPM measurement unit. Connect the plug at the site of lighter on the dashboard.

* Connecting to the battery

Connect the cable (2) to the input RPM measurement unit BAT and use tweezers to connect to the terminals of the battery, taking care not to reverse the polarity.

NOTE:

The connection of the DC voltage of the vehicle has two functions: to feed the electronic card loaded in the unit of measure and make the measurement of the regime. It is mandatory to connect the clamps on the battery test vehicle and not on a battery attached.



6.5. Measurement

During mode selection scheme using the key . Select mode regime called battery. After three seconds, the display message asks the operator the number of cylinders of the vehicle. The number of cylinders is the number of cylinders for a 4-stroke engine. You can measure a system in an engine 2 times see the following table of equivalence.

Número de cilindros	
12	
10	
8	4
6	
5	
4	2
3	
2	1
1	
4 tiempos 2 tiempos	

The revolutions account has an automatic detection system. In case of loss of the regime, each power or when the number of cylinders is changed, the unit starts to search for a stable regime (Ind.). This period can last several seconds. When the engine speed is detected, the current measurement is read directly.

In case of problems or errors of measurement:

On some vehicles, the measurement may be unstable. This phenomenon may be due to several reasons. In case of problems, (unstable measurement or inaccurate) ensure minimize the risk of error, as follows:

- The battery should be slightly discharged. The result will be better if the vehicle's headlights are left on a few minutes before starting the engine,
- During measurement, turn on the headlights to promote electricity consumption,
- If the measurement is made in making lighter, try the measurement directly on the battery,

If the value of idling, after initialization, is stable but false (double value of the real system for example), change the selection of the number of cylinders until the appropriate value appears.

7 Message from equipment.

7.1. Messages

Next messages could be presented:

Number Error	Message Displayed
1	Presence of leakage
2	Data not valid
3	HC off limits
4	CO out of bounds
5	CO2 out of bounds
6	O2 outside the boundaries
7	Nox off limits
8	Flow too weak
9	Out of paper
10	open gate
11	PB PRINT
12	HC residue gas inlet
13	Com Error. MB
14	necessary adjustment
15	Change captor O2
16	Condensation: Waiting
17	Warnming problem

7.2. Leaks

Re-testing, disconnecting the probe of the device and plugging the inlet gases decanter. If the problem disappears, the leak occurs in the probe tube.

Otherwise, it is related to the device. Call Technical service.

7.3. Invalid data

Adjustment needed. Call technical service

7.4. HC out of range

Overcoming the range.

Check that the ambient air is not polluted and redo zero.

Back to make an adjustment if the problem persists.

7.5. CO out of range

Overcoming the range.

Check that the ambient air is not polluted and redo zero.

Back to make an adjustment if the problem persists.

7.6. CO₂ out of range

Overcoming the range.

Check that the ambient air is not polluted and redo zero.

Back to make an adjustment if the problem persists.

7.7. O₂ out of range

Overcoming the range.

Check that the ambient air is not polluted and redo zero.

Check the oxygen sensor.

7.8. NO_x out of range

Overcoming the range.

Call technical service.

7.9. Flow too weak

The message flow too weak disappears when the pressure returns to normal for 45 seconds.

Verifying that flexible at the rear of the device are connected as depicted in the drawing of the back of the unit. Verify that are not folded or twisted or plugged.

Disconnect the sampling probe to see if the problem is related to this or persists.

1) If the probe is disconnected, the message no longer appears:

Check that the sampling tube is not crushed by an external mass.

Clean the probe and the flexible sampling with compressed air.

2) If the probe is disconnected, the message is always active:

Control the flow using a flow meter.

If this flow is between the minimum flow rate and the nominal flow, it is necessary for maintaining open the device for controlling the pressure switch, eventually calibrated using a pressure sensor with a depression activation at 160 mbar, or even replace it.

However, if the flow is outside the acceptable range call technical service.

7.10. Not paper.

Place a new roll of thermal printer.

7.11. Open lever.

The thermal printer lever is lowered.

7.12. PB IMPR

Printing problem, point the associated error code and call technical service..

7.13. HC residue in the inlet gas.

When a measurement ambient air is performed, if for over twenty seconds, the rate of CO₂ is less than 2% by volume while measuring the HC indicates more than 20 ppm, it is considered that there is present oil residues on the circuit input.

These wastes may also be in ambient air, so it is necessary to ensure that the workspace is properly ventilated. It may be necessary to clean the probe. If the problem persists, clean the decanter.

If the measurement of HC decreases to a rate below 20 ppm for more than twenty seconds, this message disappears and a zero is performed after the message appears ongoing measurement..

7.14. Error com. MB (Microbench)

Check that the appliance is grounded. The extensions for food should have a mandatory ground, like the socket.

Communication between the main board and bench measurement is defective. Check first that the power cables and communication are properly connected. If the problem persists call service..

7.15. Necessary adjustment

Call technical service, you need to make an adjustment to a standard gas. This message indicates a too important instrument drift.

7.16. Change O₂ sensor

During a reset, the system associates the rate of oxygen in air (20.9%) to the corresponding terminal voltage of the oxygen sensor. This tension must be enclosed in -2,15V and -1,95V. The message "change O₂ sensor" means faulty oxygen sensor or bad connection. If the voltage is not within that range will appear.

Call technical service to proceed to replace the sensor.

7.17. Condensation: wait

It may happen that this message appears during the ignition phase. You just need to wait for the appliance heat so it's ready to take measurements.

However, if this message appears during a measurement and persists, it is necessary to control the following::

- Check that the decanter is properly connected.
- Check the flow through the output connector of the condensates.
- Check the internal parameters in the maintenance menu.
- Make a leak test

Depending on the results, call Customer Service will be responsible to replace the defective part.

7.18. Warming problem.

If the estimated time of launch of the appliance is greater than 15 minutes (time between the moment the device and the time the measurements are made lights time), this message appears.

If the temperature is below 2 ° C, it is necessary to put the device in a warmer place and back on..

7.19. Other:

a) Range out of limits

On the screen, is translated as "----".

If this message appears, it means that the measurement system exceeds 9999 rpm. Check that the System menu has chosen the right taster and information relating to controlled engine are correct and which were introduced..

b) Temperature out of limits

On the screen, is translated as "----".

If the ambient temperature is below 0 ° C or if you are making a current measurement in a vehicle, this means that the measured quantity is out of range 0-150 ° C.

If not correspond to one of these cases, check that the oil temperature sensor is properly connected to the connector that is assigned. If this is not the case, replace the sampling probe.